

YATRA — AI Historical Walkthroughs

RAG + World State Integration Testing Guide

A reference document for testing retrieval accuracy, narrative generation, and image prompts.

1. System Architecture Overview

Yatra uses a unified RAG & LLM pipeline running inside the main FastAPI backend. When a walkthrough is requested, the system performs a localized search to ground the ensuing scene and prompt generation in historical fact.

Step	Component	Operation	Performance
1	API Endpoint	POST /api/walkthrough receives (place, era)	HTTP Request
2	Fast RAG Path	Checks key in chunks/snapshot_index.json	O(1) memory lookup (~0ms)
3	Fuzzy RAG Fallback	Chroma DB query (gemini-embedding-001)	Similarity search (<200ms)
4	World State Gen	Sends RAG context + prompt to Gemini API	LLM Generation (2 - 5s)

2. Server Setup & Launch

Ensure you have your environment configured inside **backend/.env** with a valid API key:

```
GEMINI_API_KEY=AIzaSyYourKeyHere...
```

Start the server from the repository root directory **Yatra/**:

```
cd /Users/dakshsablok/Yatra
PYTHONPATH=. backend/venv/bin/uvicorn backend.main:app --reload --port 8000
```

3. Integration Test Scenarios

Scenario A: Fast Path Lookup (Valid Place & Era)

Verify O(1) retrieval match works for indexed places (e.g. Hampi, Agra, Mohenjo-daro). This hits the cached JSON index instantly without querying Chroma DB.

```
curl -X POST http://localhost:8000/api/walkthrough \
  -H "Content-Type: application/json" \
  -d '{
    "place": "Hampi",
    "era": "Vijayanagara Empire (c. 1500 CE)"
  }'
```

Scenario B: List Supported Combinations

Call the helper route to list all 25 valid place/era keys stored in the database.

```
curl http://localhost:8000/api/walkthrough/places
```

4. World State Validation Criteria

When verifying the JSON response payload, your teammate must ensure the following schema constraints and architectural rules are fulfilled:

- **Stops Count:** Must generate between 3 and 5 stops.
- **Narration Word Count:** Each stop's `narration_script` must contain at least 120 words.
- **Context Grounding:** The `daily_life_facts` array must have at least 3 factual details extracted from retrieval context.
- **Image Prompt Continuity:** Every stop's `image_prompt` after the first must explicitly refer to the previous stop's vantage point or contain transition markers (e.g., 'Continuing from...', 'Leaving the...').

5. Visual Prompt Guidelines for Imagen Integration

The generated image prompts are optimized specifically for historical accuracy and architectural environments. Ensure your image generation modules (Imagen/DALL-E) adhere to these prompt qualities:

- **Environment-Focused:** Prompt details must describe the cityscape, lighting, weather, building materials, and street layout, rather than close-up characters or modern tourist settings.
- **Material Grounding:** Expect prompts to contain specific historical materials (e.g., 'sandstone', 'terracotta brick', 'monolithic granite carvings', 'bitumen sealants') derived from the RAG database.
- **Transition Prompts:** Prompts will naturally reference previous stops to preserve camera continuity, facilitating an immersive 'walkthrough' progression.